Whole cell lysates from tissue culture cells

Whole cell lysates are relatively simple and rapid to generate from adherent and suspension tissue culture cells, or even tissue slices, when a biochemical activity or cellular component cannot be easily examined in intact cells (in cellulo) or living organisms (in vivo) or needs to be reconstituted or interrogated in vitro.

Success often depends on detergent composition and stringency of the lysis buffer. RIPA lysis buffer (*Alternative A*) is a strong lysis buffer suitable for extracting a wide range of proteins from various cellular compartments. NP-40 lysis buffer (*Alternative B*) is milder and used primarily to extract cytoplasmic proteins under non-denaturing conditions. Urea lysis buffer (*Alternative C*) is optimized for proteome analysis by mass spectrometry.

While many researchers assume that their chosen method for total protein extraction faithfully reflects the specimen's protein composition, few rigorously compare different protocols to validate this assumption. Moreover, whole cell lysis inherently destroys subcellular compartmentalization, which can lead to artifactual interactions between proteins that are normally separated. Consequently, findings obtained from whole cell lysates should be critically evaluated. Where compartmentalization matters, isolate and fractionate first. Lysates destined for immunoprecipitation continue with the protocol for soluble proteins SOP 0027.

Risk assessment

- – Work with human-derived material or transgenic cell lines (BSL-2)
- ▷ Wear gloves, safety glasses, lab coat
- ◻ Dispose waste after inactivation as REGULATED MEDICAL WASTE

Reviewed: May 22, 2026

Procedures

›› *Choosing a suitable lysis buffer*

(1.) Select a lysis buffer based on the target protein class and intended downstream application. The table below summarizes the solubility of representative protein classes in each buffer.

Protein source	RIPA	NP-40	Other extraction methods or buffers
Whole cell extracts	Good	Recommended	
Cytosolic proteins	Good	Good	Tris-HCl
Nuclear proteins	Recommended	Limited	Nuclear extraction SOP 0025
Mitochondrial proteins	Recommended	Limited	
Membrane-bound proteins	Recommended	Good	
High molecular weight proteins	Poor	Poor	Urea
Structural proteins: Fibronectin, Keratin, Lamin, Tubulin	Limited	Poor	Tris-Triton
Signaling proteins: EGFR, HSP90, c-Src	Poor	Poor	
Apoptotic markers: Cleaved Caspase-8	Insoluble	Insoluble	
Transcription factors: GATA2	Insoluble	Insoluble	Nuclear extraction SOP 0025
Heterochromatin: HP1, H3K9me3	Insoluble	Insoluble	MNase digestion SOP 0025
Histones: H3, H4	Insoluble	Insoluble	Acid extraction SOP 0026

Hint: In cases where protein solubility is poor, the specimen can be taken up directly in Laemmli sample buffer to analyze protein recovery by immunostaining. However, the high SDS content of this buffer interferes with other applications such as immunoprecipitation of protein complexes and most enzymatic assays.

›› *Preparation of cell lysates*

- ☐ Phosphate-buffered saline, pH 7.4 R0037
- ☐ 100 × Inhibitor cocktail R0090
- ☐ Cell scraper, plastic
- ☐ Microcentrifuge tube, round-bottom
- ☐ Sonicator (*optional*)

(1.) Start with $1{\times}10^6$–$1{\times}10^9$ cells, depending on cell type, or with a weighed tissue specimen.

> ***Note:*** Both fresh and frozen cell pellets can be used. There is no need to thaw frozen cell pellets.

> ***Critical:*** Keep buffer volumes to a minimum to maintain high protein concentrations.

(2.) Wash the cells twice with ice-cold PBS containing the desired inhibitors and bring the material into lysis buffer, depending on the specimen:

- Suspension cells: collect at 400 × *g* for 5 min at 4 °C after each wash, then resuspend the pellet in lysis buffer.
- Adherent cells: aspirate the last wash completely, apply lysis buffer directly to the dish, scrape with a cold plastic cell scraper, and transfer the lysate to a fresh tube.
- Tissue: weigh the specimen, cut it into pieces on ice in a round-bottom microcentrifuge tube, snap-freeze in liquid nitrogen, and homogenize in lysis buffer on ice.

(3.) If the lysate is viscous or membrane and nuclear proteins extract poorly, reduce the viscosity and aid solubilization by removing nucleic acids in one of the following ways:

- Sonicate the lysate twice for 10 s at 20 kHz on ice to disrupt cellular membranes and shear genomic DNA.

  > ***Critical:*** Since sonication can damage proteins, minimize pulse length and number of pulses. Apply short pulses and lower the power level to avoid frothing.

- Add a universal nuclease at 25 U per 1 mL of lysate and incubate for 15 min at room temperature.

  > ***Critical:*** The nuclease requires magnesium, and the RIPA and NP-40 lysis buffers carry 1 mM EDTA. Supplement the lysate with $MgCl_2$ to 2 mM before adding the enzyme.

A › *Protein extraction with RIPA lysis buffer*

- ☐ Radioimmunoprecipitation lysis buffer, pH 7.6 R0148

(1.) Add 50–100 vol ice-cold 1 × RIPA lysis buffer with inhibitors, about 1 mL per $1{\times}10^7$ cells or 5 mg of tissue. Agitate the suspension for 20 min at 4 °C. 20 min

> ***Critical:*** Tissue samples will take 1–2 h or longer to complete lysis. Homogenize with an electric homogenizer or mortar.

(2.) Remove the insoluble fraction by centrifugation at 12 000 × *g* for 20 min at 4 °C. 20 min

(3.) Carefully collect the supernatant containing the soluble protein fraction in a new tube on ice.

B › *Protein extraction with NP-40 lysis buffer*

- ☐ NP-40 lysis buffer, pH 7.4 R0149
  – *or*: Tris-Triton buffer, pH 7.4 R0150

(1.) Add 50–100 vol ice-cold 1 × NP-40 lysis buffer with inhibitors, about 1 mL per $1{\times}10^7$ cells or 5 mg of tissue. Agitate the suspension for 20 min at 4 °C. 20 min

> ***Critical:*** Tissue samples will take 1–2 h or longer to complete lysis. Homogenize with an electric homogenizer or mortar.

(2.) Remove the insoluble fraction by centrifugation at 12 000 × *g* for 20 min at 4 °C. 20 min

(3.) Carefully collect the supernatant containing the soluble protein fraction in a new tube on ice.

C › *Proteome extraction*

☐ Urea lysis buffer, pH 7.4 R0151

(1.) Add 5 vol ice-cold 1 × urea lysis buffer with inhibitors, about 100 µL per $1{\times}10^7$ cells or 5 mg of tissue. Pipette the sample up and down to break up cell clumps and vortex gently.

(2.) Boil the lysate at 95 °C for 5 min.

(3.) *Optional:* Sonicate the lysate twice for 10 s at 20 kHz on ice to disrupt cellular membranes, shear genomic DNA, and aid with protein solubilization.

> ***Critical:*** Since sonication can damage proteins, minimize pulse length and number of pulses. Apply short pulses and lower the power level to avoid frothing.

> ***Hint:*** Alternatively, use a universal nuclease to enzymatically digest DNA and RNA. Use 25 U per 1 mL of cell lysate; incubate for 15 min at room temperature.

(4.) Remove the insoluble fraction by centrifugation at 12 000 × *g* for 20 min at 4 °C. 20 min

(5.) Carefully collect the supernatant containing the soluble protein fraction in a new tube on ice.

Analyses

- Determine the total protein concentration against 1 × lysis buffer.
- Prepare samples for SDS-PAGE SOP 0007 corresponding to $2{\times}10^5$ cells (or 20 µg protein) for the soluble and for the insoluble fraction of the whole-cell extract. Immunostain for the protein of interest.

Materials

Preparation of cell lysates

Phosphate-buffered saline R0037, pH 7.4

Inhibitor cocktail R0090, 100 ×

Cell scraper, plastic

Microcentrifuge tube, round-bottom

Sonicator (*optional*)

Protein extraction with RIPA lysis buffer

Radioimmunoprecipitation lysis buffer R0148, pH 7.6, Store at 4 °C. Alternatively, store at −20 °C.

Protein extraction with NP-40 lysis buffer

NP-40 / Tris-Triton lysis buffer

- **NP-40 lysis buffer** R0149, pH 7.4, Store at 4 °C. Alternatively, store at −20 °C.
- *or* **Tris-Triton buffer** R0150, pH 7.4, Store at 4 °C. Alternatively, store at −20 °C.

Proteome extraction

Urea lysis buffer R0151, pH 7.4, Store at . To avoid precipitation.

Recipes

Radioimmunoprecipitation lysis buffer (RIPA), pH 7.6

Amount	Ingredient		Stock	Final
2.5 mL	Tris hydrochloride (Tris-Cl), pH 7.4	R0055	1 M	50 mM
1.5 mL	Sodium chloride (NaCl)	R0046	5 M	150 mM
500 µL	IGEPAL® CA-630	[9002-93-1]	100%	1.0%
250 mg	Sodium deoxycholate	[302-95-4]	414.55 g/mol	0.5%
250 µL	Sodium dodecylsulfate (SDS)	R0047	20%	0.1%
	Skin irritation (H315); Serious eye damage (H318)			
100 µL	EDTA, pH 8.0	R0017	0.5 M	1 mM
100 µL	Dithiothreitol (DTT)	☐ R0015	1 M	2 mM
	Skin irritation (H315); Serious eye irritation (H319)			
To 50 mL	Water, reagent-grade			

Filter sterilize. Add reducing agent right before use. Store at 4 °C. Alternatively, store at −20 °C. ***Critical:*** Due to limited solubility, this buffer cannot be prepared as concentrated stock. ***Note:*** This is Proteintech's optimized RIPA buffer recipe. ***This is why:*** IGEPAL® CA-630 is a mild detergent alternative to Nonidet P-40 which is no longer manufactured; it can be replaced with Triton™ X-100.

Radioimmunoprecipitation lysis buffer (RIPA)

50 mM Tris-Cl, 150 mM NaCl, 1.0% IGEPAL® CA-630, 0.5% Sodium deoxycholate, 0.1% SDS, 1 mM EDTA, ☐ 2 mM DTT, pH 7.6

No GHS hazards at the indicated concentrations

☐ Hold for EHS review before disposal

Date: Sign: R0148

NP-40 lysis buffer, pH 7.4

Amount	Ingredient		Stock	Final
2.5 mL	Tris hydrochloride (Tris-Cl), pH 7.4	R0055	1 M	50 mM
1.5 mL	Sodium chloride (NaCl)	R0046	5 M	150 mM
500 µL	IGEPAL® CA-630	[9002-93-1]	100%	1.0%
100 µL	Dithiothreitol (DTT)	☐ R0015	1 M	2 mM
	Skin irritation (H315); Serious eye irritation (H319)			
100 µL	EDTA, pH 8.0	R0017	0.5 M	1 mM
To 50 mL	Water, reagent-grade			

Filter sterilize. Add reducing agent right before use. Store at 4 °C. Alternatively, store at −20 °C. ***Note:*** Add up to 1% sodium deoxycholate to increase stringency. ***This is why:*** IGEPAL® CA-630 is a mild detergent alternative to Nonidet P-40 which is no longer manufactured; it can be replaced with Triton™ X-100.

NP-40 lysis buffer

50 mM Tris-Cl, 150 mM NaCl, 1.0% IGEPAL® CA-630, ☐ 2 mM DTT, 1 mM EDTA, pH 7.4

No GHS hazards at the indicated concentrations

☐ Hold for EHS review before disposal

Date: Sign: R0149

Tris-Triton buffer, pH 7.4

Amount	Ingredient		Stock	Final
500 µL	Tris hydrochloride (Tris-Cl), pH 7.4	R0055	1 M	10 mM
1.0 mL	Sodium chloride (NaCl)	R0046	5 M	100 mM
5.0 mL	Triton™ X-100	R0057	10%	1.0%
	Skin irritation (H315); Serious eye damage (H318); Aquatic toxicity (long-term) (H411)			
250 mg	Sodium deoxycholate	[302-95-4]	414.55 g/mol	0.5%
250 µL	Sodium dodecylsulfate (SDS)	R0047	20%	0.1%
	Skin irritation (H315); Serious eye damage (H318)			
10 mL	Glycerol	R0022	50%	10%
100 µL	Dithiothreitol (DTT)	☐ R0015	1 M	2 mM
	Skin irritation (H315); Serious eye irritation (H319)			
To 50 mL	Water, reagent-grade			

Filter sterilize. Add reducing agent right before use. Store at 4 °C. Alternatively, store at −20 °C.

Tris-Triton buffer

10 mM Tris-Cl, 100 mM NaCl, 1.0% Triton™ X-100, 0.5% Sodium deoxycholate, 0.1% SDS, 10% Glycerol, ☐ 2 mM DTT, pH 7.4

Aquatic toxicity (long-term)

☐ Hold for EHS review before disposal

Date: Sign: R0150

Urea lysis buffer, pH 7.4

Amount	Ingredient		Stock	Final
10 mL	HEPES, pH 7.5	R0024	1 M	200 mM
27.0 g	Urea, ultrapure	[57-13-6]	60.06 g/mol	9 M
To 50 mL	Water, reagent-grade			

Reconstitute urea in a water bath of 25–30 °C for 30 min to aid dissolution. Store at . To avoid precipitation. ***Critical:*** Do not warm above 30 °C since isocyanates will form and irreversibly modify proteins! Prepare fresh, use within one day.

Urea lysis buffer

200 mM HEPES, 9 M Urea, pH 7.4

No GHS hazards at the indicated concentrations

- ☐ Dilute below the discharge limit
- ☐ Flush to drain with excess water

Date: Sign: R0151

Troubleshooting

Choosing a suitable lysis buffer

In Step 1:

- Target protein not detected in the soluble fraction
  - Check the insoluble pellet by resuspending directly in Laemmli sample buffer. If the target is present in the pellet, increase detergent stringency or switch to a urea-based buffer.

Preparation of cell lysates

In Step 2:

- Low protein yield
  - Verify cell count and ensure sufficient lysis buffer volume. Use at least 50–100 packed-cell volumes of buffer.
  - Extend lysis time or add sonication to improve solubilization of membrane and nuclear proteins.
- Protein degradation visible as unexpected low-MW bands or loss of signal
  - Add fresh protease inhibitor cocktail immediately before lysis. Keep all steps at 4 °C. Ensure samples are not left on ice for extended periods without inhibitors.

Change log

2020-03-22	Shany Koren-Hauer	Initial version.
2025-02-22	Benjamin C. Buchmuller	Adaptation as SOP; scope extended beyond RIPA to NP-40 and Tris-Triton, informed by Invent Biotechnologies "RIPA Buffer: The Potential Troubles" mini-review.
2026-03-03	Benjamin C. Buchmuller	Corrected volumes for final DTT concentration.
2026-05-22	Benjamin C. Buchmuller	Corrected volumes for Tris-Triton buffer Triton X-100 (500 µL → 5.0 mL; v1 prepared at 0.1% vs intended 1.0%, significantly weaker detergent for cytoskeletal extraction) and Glycerol (5.0 mL → 10 mL; v1 prepared at 5% vs intended 10%, mildly less protein stabilization). Same shape as the prior DTT correction.